General Mathematics — Exercise Sheet 9

Implicit derivatives and mean value theorems

Variant: Statements

Exercise 1 (Two derivatives of a given branch). Suppose a twice differentiable function y is given on an open interval containing 2, with $y(x) > 0$, $y(2) = 1$, and

$$xy(x) + \ln y(x) = 2.$$

Find $y'(2)$ and $y''(2)$ by differentiating the identity. State the nonzero coefficient that allows each derivative to be solved for at that point. No branch-existence theorem is assumed.

Exercise 2 (An exponential parameter). For $t \in \mathbb{R}$, let $x(t) = e^t$ and $y(t) = te^t$. Find the graph derivatives dY/dx and d^2Y/dx^2 in terms of t. Find the tangent line at $t = \ln 2$. Eliminate the parameter and verify both derivative formulas from the resulting graph. Use Lecture 9's parametric formulas and Lecture 8's elementary rules.

Exercise 3 (Rolle's theorem, monotonicity and constancy).

1. Apply Rolle's theorem to $p(x) = x^4 - 2x^2$ on $[-\sqrt{2}, \sqrt{2}]$. Find every interior point at which p' is zero and determine the monotonicity intervals. Explain why several zero-derivative points are consistent with Rolle's theorem.
2. Prove by the zero-derivative theorem that $H(x) = \arcsin x + \arccos x$ is constant on $[-1, 1]$, and find its value. Check the endpoint hypotheses although the derivative formulas hold only inside.

Exercise 4 (A residual bounds the error in a root). Recall from Sheet 8 that $f(x) = e^x + x$ is a continuous strictly increasing bijection of $\mathbb{R}$ and $f'(x) = e^x + 1$. Let r be the unique root of $f(r) = 5$. Use the mean value theorem to prove

$$|z - r| \le |f(z) - 5| \qquad (z \in \mathbb{R}).$$

For the trial input $z = \ln 4$, prove $1 < r < \ln 4$ and the sharper estimate

$$0 < \ln 4 - r < \frac{\ln 4 - 1}{e + 1}.$$

Use the established bound $2 < e < 3$.

Exercise 5 (Leibniz's formula for higher derivatives). For integers $n \geq 1$, define $n! = 1 \cdot 2 \cdots n$, and set $0! = 1$. For $0 \leq k \leq n$, write $\binom{n}{k} = n!/(k!(n-k)!)$. Suppose f and g have derivatives through order n on an open interval. With $f^{(0)} = f$ and $g^{(0)} = g$, prove by induction the formula

$$(fg)^{(n)} = \sum_{k=0}^{n} \binom{n}{k} f^{(k)} g^{(n-k)}.$$

The sum notation denotes a finite sum over the printed indices. Use the result to calculate the nth derivative of $x^2 e^x$ for every integer $n \geq 0$.